Pearson Edexcel A-Level Mathematics (9MA0)

Module 2

Functions, Expansions & Advanced Algebra

Paper 1 & 2 (Pure) · 90 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
2.1 Algebraic Methods	27	____ / 27
2.2 Functions and Graphs	24	____ / 24
2.3 Binomial Expansion	18	____ / 18
2.4 Exponentials and Logarithms	21	____ / 21

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

2.1 Algebraic Methods

Proof (deduction, exhaustion, contradiction, counter-example), algebraic fractions, polynomial division, the factor theorem and partial fractions.

Key facts

- Even: $2n$; odd: $2n+1$; consecutive: $n, n+1$
- Factor theorem: $f(a)=0 \Leftrightarrow (x-a)$ is a factor; $f\left(\frac{b}{a}\right)=0 \Leftrightarrow (ax-b)$ is a factor
- $\frac{px+q}{(x-a)(x-b)} \equiv \frac{A}{x-a}+\frac{B}{x-b}$
- Repeated factor: $\frac{\ldots}{(x-a)^2(x-b)} \equiv \frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
- Improper fraction (degree top $\geq$ degree bottom): divide first

Method

- Proof by contradiction: 'Assume the opposite...', reach an impossibility, then conclude.
- Partial fractions: multiply by the denominator, substitute the roots, compare coefficients for the rest.
- Proofs must end with a conclusion sentence that answers the question.

Exam traps

- Proof by example is not a proof.
- Not stating the assumption explicitly in a contradiction proof.
- Forgetting the $\frac{B}{(x-a)^2}$ term for a repeated factor.

In the exam: A proof question appears on nearly every Paper 1/2 (4-5 marks). Partial fractions feed into binomial expansion and integration.

Specification checklist

Skill	Learnt	Practised	Mastered
Proof by deduction	☐	☐	☐
Proof by exhaustion	☐	☐	☐
Proof by contradiction	☐	☐	☐
Disproof by counter-example	☐	☐	☐
Algebraic fractions	☐	☐	☐
Polynomial division	☐	☐	☐
Factor theorem	☐	☐	☐
Partial fractions	☐	☐	☐
Partial fractions with repeated factors	☐	☐	☐

Videos (click to open)

☐ **Recap** — Proof (Year 1) in less than 18 minutes • A-Level Maths, Pure Year 1, Chapter 7b (Bicen Maths)
☐ **Recap** — Proof by Contradiction in less than 18 minutes • A-Level Maths, Pure Year 2, Chapter 1a (Bicen Maths)
☐ **Recap** — Factor Theorem in less than 18 minutes • A-Level Maths, Pure Year 1, Chapter 7a (Bicen Maths)
☐ Proof by deduction — Introducing Proof by Deduction (TLMaths)
☐ Proof by exhaustion — Introducing Proof by Exhaustion (TLMaths)
☐ Proof by contradiction — Introducing Proof by Contradiction (TLMaths)
☐ Disproof by counter-example — Disproof by Counter Example Examples (TLMaths)
☐ Algebraic fractions — Rational Expressions: Examples of Simplifying Algebraic Fractions (TLMaths)
☐ Polynomial division — Polynomials: Polynomial Division using the Long Division Method (TLMaths)
☐ Factor theorem — Polynomials: Introducing the Factor Theorem (TLMaths)
☐ Partial fractions — Algebraic Fractions: Introducing Partial Fractions via Substitution (TLMaths)
☐ Partial fractions with repeated factors — Algebraic Fractions: How we deal with Repeated Factors in the Denominator (TLMaths)

Q2.1.01 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{35-2x}{x^2+x-20}$ in partial fractions.

(3)

WORKING

Q2.1.02 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{3x^2-6x-9}{x^2+6x+5}$.

(3)

WORKING

Q2.1.03 Proof by exhaustion ●○○ [2 marks] ☐ Todo ☐ Mastered

Prove that for all integers n with $1 \leq n \leq 6$, n^2+2 is not divisible by 4.

(2)

WORKING

Q2.1.04 Disproof by counter-example ●○○ [2 marks] ☐ Todo ☐ Mastered

Disprove: 'if $a > b$ then $a^2 > b^2$ for all real a, b'.

(2)

WORKING

Q2.1.05 Polynomial division ●○○ [4 marks] ☐ Todo ☐ Mastered

Divide $x^3 + 7x^2 + 4x - 12$ by $(x - 1)$ and hence factorise $x^3 + 7x^2 + 4x - 12$ completely.

(4)

WORKING

Q2.1.06 Proof by deduction ●○○ [3 marks] ☐ Todo ☐ Mastered

Prove that the sum of any three consecutive integers is divisible by 3.

(3)

WORKING

Q2.1.07 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{28}{x^2 - x - 12}$ in partial fractions.

(3)

WORKING

Q2.1.08 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{x^2 - 3x + 2}{x^2 - 5x + 6}$.

(3)

WORKING

Q2.1.09 **Factor theorem** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

$f(x) = 3x^3 + x^2 + 2x + 4$.

Use the factor theorem to determine whether $(x - 1)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.10 **Proof by exhaustion** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Prove that the square of any integer is either a multiple of 3 or one more than a multiple of 3.

(4)

WORKING

Q2.1.11 **Polynomial division** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Divide $2x^3 - 16x^2 + 38x - 24$ by $(x - 4)$ and hence factorise $2x^3 - 16x^2 + 38x - 24$ completely.

(4)

WORKING

Q2.1.12 Proof by deduction ●○○ [2 marks] ☐ Todo ☐ Mastered

Prove that $(n+3)^2 - (n-3)^2$ is a multiple of 12 for all integers n.

(2)

WORKING

Q2.1.13 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{7x-11}{x^2-x-6}$ in partial fractions.

(3)

WORKING

Q2.1.14 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{x^2+5x+6}{x^2-3x-18}$.

(3)

WORKING

Q2.1.15 Factor theorem ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = 2x^3 - 4x^2 + 6x + 3$.

Use the factor theorem to determine whether $(x-2)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.16 Polynomial division ●○○ [4 marks] ☐ Todo ☐ Mastered

Divide $x^3 - 6x^2 - 7x + 60$ by $(x + 3)$ and hence factorise $x^3 - 6x^2 - 7x + 60$ completely.

(4)

WORKING

Q2.1.17 Proof by deduction ●○○ [3 marks] ☐ Todo ☐ Mastered

Prove that $x^2 - 6x + 11 > 0$ for all real values of x.

(3)

WORKING

Q2.1.18 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{17 - 4x}{x^2 - 9x + 20}$ in partial fractions.

(3)

WORKING

Q2.1.19 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{2x^2 - 14x + 24}{x^2 + x - 12}$.

(3)

WORKING

Q2.1.20 Factor theorem ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - 4x^2 + 2x - 9$.

Use the factor theorem to determine whether $(x - 1)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.21 Proof by contradiction ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove by contradiction that there are infinitely many prime numbers.

(5)

WORKING

Q2.1.22 Factor theorem ●●○ [6 marks] ☐ Todo ☐ Mastered

$f(x) = 2x^3 + 18x^2 + px + 48$, where p is a constant.

Given that $(x + 2)$ is a factor of $f(x)$,

(a) find the value of p, **(2)**

(b) hence factorise $f(x)$ completely. **(4)**

WORKING

Q2.1.23 Partial fractions with repeated factors ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\frac{-x^2+7x+13}{(x-3)(x+2)^2} \equiv \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{x-3}$, find the values of A, B and C.

(4)

WORKING

Q2.1.24 Proof by contradiction ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove by contradiction that $\sqrt{2}$ is irrational.

(5)

WORKING

Q2.1.25 Partial fractions with repeated factors ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\frac{-x^2+7x-14}{(x-3)^2(x-1)} \equiv \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{x-1}$, find the values of A, B and C.

(4)

WORKING

Q2.1.26 **Partial fractions with repeated factors** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Given that $\frac{7x^2+23x-34}{(x-3)(x+4)^2} \equiv \frac{A}{x+4} + \frac{B}{(x+4)^2} + \frac{C}{x-3}$, find the values of A, B and C.

(4)

WORKING

Q2.1.27 **Proof by contradiction** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Prove by contradiction that if n^2 is even then n is even.

(4)

WORKING

MY NOTES — 2.1 ALGEBRAIC METHODS: MISTAKES TO REMEMBER

2.2 Functions and Graphs

Functions: domain/range, composite and inverse functions, the modulus function, and transformations of graphs.

Key facts

- $fg(x) = f(g(x))$: apply g first
- f^{-1} exists only if f is one-to-one; domain of f^{-1} = range of f; $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$
- $y = f(x) + a$: up a; $y = f(x + a)$: left a; $y = af(x)$: vertical stretch s.f. a; $y = f(ax)$: horizontal stretch s.f. $\frac{1}{a}$
- $y = |f(x)|$: reflect negative parts up; $y = f(|x|)$: reflect the right-hand half in the y-axis

Method

- Solve $|f(x)| = g(x)$ by sketching both graphs, then solve $f(x) = g(x)$ and $-f(x) = g(x)$ and check each answer.
- To find an inverse: write $y = f(x)$, rearrange for x, swap letters.
- Combined transformations: do changes inside the bracket (horizontal) and outside (vertical) separately.

Exam traps

- $f^{-1}(x) \neq \frac{1}{f(x)}$.
- $f(x + 2)$ moves the graph left, not right.
- Not rejecting modulus solutions that make the right-hand side negative.

In the exam: Functions questions are long (8-12 marks), mixing range, composite, inverse and a modulus sketch.

Specification checklist

Skill	Learnt	Practised	Mastered
Modulus function y = |x|	☐	☐	☐
Solving modulus equations and inequalities	☐	☐	☐
Mappings and functions, domain and range	☐	☐	☐
Composite functions	☐	☐	☐
Inverse functions	☐	☐	☐
Transformations y = af(x), f(x)+a, f(x+a), f(ax)	☐	☐	☐
Modulus graphs y = |f(x)| and y = f(|x|)	☐	☐	☐
Combining transformations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Graphs & Transformations in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 4 (Bicen Maths)
☐ **Recap** — Functions & Graphs in less than 26 minutes • A-Level Maths, Pure Year 2, Chapter 2 (Bicen Maths)
☐ Modulus function y = |x| — Graphs: Introducing the Modulus Function (TLMaths)
☐ Solving modulus equations and inequalities — Graphs: Introducing Solving Modulus Equations (TLMaths)
☐ Mappings and functions, domain and range — Functions: Examples of Finding the Domain and Range (TLMaths)
☐ Composite functions — Functions: The Domain of a Composite Function (TLMaths)
☐ Inverse functions — Functions: Finding an Inverse Function (TLMaths)
☐ Transformations y = af(x), f(x)+a, f(x+a), f(ax) — Graph Transformations: Beginning an Investigation into Transformations (TLMaths)
☐ Modulus graphs y = |f(x)| and y = f(|x|) — Graphs: EDEXCEL Sketching y=f(|x|) (TLMaths)
☐ Combining transformations — Trigonometry | Combining Transformations of Trig Graphs y=f(ax+b) types (ExamSolutions)

Q2.2.01 Modulus function y = |x| ●○○ [4 marks] ☐ Todo ☐ Mastered

Sketch $y = |x + 4| - 1$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.02 Solving modulus equations and inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $|x + 5| < 5$.

(3)

WORKING

Q2.2.03 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 + 8x + 25$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.04 Composite functions ●○○ [6 marks] ☐ Todo ☐ Mastered

The functions f and g are defined by $f(x) = -2x - 2$, $x \in \mathbb{R}$ and $g(x) = 2x^2 + 5$, $x \in \mathbb{R}$.

(a) Find $fg(3)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.05 Transformations y = af(x), f(x)+a, f(x+a), f(ax) ●○○ [2 marks] ☐ Todo ☐ Mastered

The curve $y = f(x)$ has a maximum point at $(-1, 1)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(-x)$, and describe the transformation.

(2)

WORKING

Q2.2.06 Modulus function y = |x| ●○○ [4 marks] ☐ Todo ☐ Mastered

Sketch $y = |x + 4| - 3$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.07 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 + 2x + 9$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.08 **Composite functions** ●○○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The functions f and g are defined by $f(x) = -x - 5$, $x \in \mathbb{R}$ and $g(x) = x^2 - 4$, $x \in \mathbb{R}$.

(a) Find $fg(2)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.09 **Transformations y = af(x), f(x)+a, f(x+a), f(ax)** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

The curve $y = f(x)$ has a maximum point at $(0, 4)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(x - 2)$, and describe the transformation.

(2)

WORKING

Q2.2.10 **Modulus function y = |x|** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Sketch $y = 2|x + 4| + 5$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.11 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 - 10x + 29$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.12 Composite functions ●○○ [6 marks] ☐ Todo ☐ Mastered

The functions f and g are defined by $f(x) = 1 - 5x$, $x \in \mathbb{R}$ and $g(x) = x^2 - 3$, $x \in \mathbb{R}$.

(a) Find $fg(-3)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.13 Transformations y = af(x), f(x)+a, f(x+a), f(ax) ●○○ [2 marks] ☐ Todo ☐ Mastered

The curve $y = f(x)$ has a maximum point at $(3, -2)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(-x)$, and describe the transformation.

(2)

WORKING

Q2.2.14 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 - 9x + 20$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.15 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{4x+3}{3x-1}$, $x \in \mathbb{R}$, $x \neq \frac{1}{3}$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.16 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = 3(x-4)^2 + 2$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

Q2.2.17 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 - 2x - 3$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.18 Solving modulus equations and inequalities ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $|2x - 6| = 2x + 1$.

(4)

WORKING

Q2.2.19 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{4 - 3x}{x - 5}$, $x \in \mathbb{R}$, $x \neq 5$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.20 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = -1(x + 2)^2 + 1$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

Q2.2.21 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 + 2x - 3$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.22 Solving modulus equations and inequalities ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $|x + 1| = -2x - 3$.

(4)

WORKING

Q2.2.23 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{3x+1}{3x-3}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.24 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = -1(x+4)^2 + 2$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

MY NOTES — 2.2 FUNCTIONS AND GRAPHS: MISTAKES TO REMEMBER

2.3 Binomial Expansion

Binomial expansion for positive integer powers (finite) and for any rational power (infinite, valid only for small x).

Key facts

- $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$
- $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$
- For rational n, $(1+bx)^n$ is valid for $|bx| < 1$, i.e. $|x| < \frac{1}{|b|}$
- $(a+bx)^n = a^n\left(1+\frac{b}{a}x\right)^n$, valid for $|x| < \frac{a}{|b|}$

Method

- Put brackets around the whole term, e.g. $(-3x)^2 = 9x^2$.
- For $(a+bx)^n$ take out a^n first, then expand.
- With partial fractions: expand each fraction separately; the overall validity is the narrower interval.

Exam traps

- Forgetting to raise a to the power n when factoring out.
- Not stating the range of validity.
- Sign errors with negative n: $(1-x)^{-1} = 1 + x + x^2 + \ldots$

In the exam: 4-6 marks; often followed by 'use your expansion to estimate...' with a substitution.

Specification checklist

Skill	Learnt	Practised	Mastered
(a + b)^n for positive integer n	☐	☐	☐
(1 + x)^n for rational n	☐	☐	☐
(a + bx)^n for rational n	☐	☐	☐
Range of validity	☐	☐	☐
Using partial fractions with expansions	☐	☐	☐
Approximations using expansions	☐	☐	☐

Videos (click to open)

☐ **Recap** — Binomial Expansion (Year 1) in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 8 (Bicen Maths)
☐ **Recap** — Binomial Expansion (Year 2) in less than 21 minutes • A-Level Maths, Pure Year 2, Chapter 4 (Bicen Maths)
☐ (a + b)^n for positive integer n — Binomial Expansion: Explaining where nCr comes from (TLMaths)
☐ (1 + x)^n for rational n — Binomial Expansion (partial fractions type) (ExamSolutions)
☐ (a + bx)^n for rational n — Binomial Expansion: Writing (a + bx)^n in the form p(1 + qx)^n (TLMaths)
☐ Range of validity — Binomial Expansion: Introducing the Range of Validity (TLMaths)
☐ Using partial fractions with expansions — Algebraic Fractions: Partial Fractions with Binomial Expansion (TLMaths)
☐ Approximations using expansions — Binomial Expansion: Approximating 1.03^8 WITHOUT a calculator (TLMaths)

Q2.3.01 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(7+4x)^{-1}$ is valid.

(1)

WORKING

Q2.3.02 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1+4x)^7$, giving each term in its simplest form.

(4)

WORKING

Q2.3.03 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(9-3x)^{\frac{1}{2}}$ is valid.

(1)

WORKING

Q2.3.04 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1-2x)^4$, giving each term in its simplest form.

(4)

WORKING

Q2.3.05 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(5+6x)^{-\frac{1}{3}}$ is valid.

(1)

WORKING

Q2.3.06 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(2+4x)^5$, giving each term in its simplest form.

(4)

WORKING

Q2.3.07 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(3-5x)^{\frac{1}{2}}$ is valid.

(1)

WORKING

Q2.3.08 Approximations using expansions ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Expand $\sqrt{1+4x}$ in ascending powers of x up to and including the term in x^2. **(3)**

(b) By substituting $x=\frac{1}{20}$, find an approximation for $\sqrt{\frac{6}{5}}$, giving your answer to 5 decimal places. **(2)**

WORKING

Q2.3.09 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1+2x)^{-\frac{3}{2}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.10 Approximations using expansions ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Expand $\sqrt{1+2x}$ in ascending powers of x up to and including the term in x^2. **(3)**

(b) By substituting $x = \frac{1}{100}$, find an approximation for $\sqrt{\frac{51}{50}}$, giving your answer to 5 decimal places. **(2)**

WORKING

Q2.3.11 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1+\frac{1}{2}x)^{-\frac{1}{3}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.12 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1-2x)^{-\frac{1}{3}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.13 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-12x-2}{1-9x^2}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.14 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(4+x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

Q2.3.15 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-11x-2}{-6x^2+x+1}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.16 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(4-3x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

Q2.3.17 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-5x-7}{-3x^2+2x+1}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.18 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(9-3x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

MY NOTES — 2.3 BINOMIAL EXPANSION: MISTAKES TO REMEMBER

2.4 Exponentials and Logarithms

Exponentials and logarithms: laws of logs, solving equations, e^x and $\ln x$, growth/decay models and reducing curves to straight lines.

Key facts

- $\log_a x = n \Leftrightarrow a^n = x$
- $\log_a(xy) = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$, $\log_a x^k = k\log_a x$
- $\frac{d}{dx}e^{kx} = ke^{kx}$, $\ln(e^x) = x$, $e^{\ln x} = x$
- $y = ab^x \Rightarrow \log y = \log a + x\log b$ (plot $\log y$ against x)
- $y = ax^n \Rightarrow \log y = \log a + n\log x$ (plot $\log y$ against $\log x$)

Method

- Solve $a^x = b$ by taking logs: $x = \frac{\ln b}{\ln a}$.
- For $e^{2x} - 5e^x + 6 = 0$ use $u = e^x$.
- In models $N = Ae^{kt}$: A is the initial value; use one data point to find k.

Exam traps

- $\log(x + y) \neq \log x + \log y$.
- Forgetting to reject $\ln$ of a negative number.
- Leaving the answer in the wrong form (exact vs 3 s.f.).

In the exam: Modelling questions (8-10 marks) are common in Paper 2; log-linear graphs appear regularly.

Specification checklist

Skill	Learnt	Practised	Mastered
Exponential functions and graphs	☐	☐	☐
e^x and ln x	☐	☐	☐
Laws of logarithms	☐	☐	☐
Solving exponential equations	☐	☐	☐
Exponential growth and decay models	☐	☐	☐
Non-linear data: log y against x and log y against log x	☐	☐	☐
Gradient of e^kx	☐	☐	☐

Videos (click to open)

☐ **Recap** — Exponentials & Logarithms in less than 32 minutes • A-Level Maths, Pure Year 1, Chapter 14 (Bicen Maths)
☐ Exponential functions and graphs — Exponentials: Sketching y = e^x (TLMaths)
☐ e^x and ln x — Logarithms: Introducing The Natural Logarithm ln(x) (TLMaths)
☐ Laws of logarithms — Laws of Logarithms: Introducing the Laws of Logarithms (TLMaths)
☐ Solving exponential equations — Exponential Equations: Solve 2^(x+3) = 3^(3-x) (TLMaths)
☐ Exponential growth and decay models — Exponential Growth & Decay: Long-Term Behaviour (TLMaths)
☐ Non-linear data: log y against x and log y against log x — Reduction to Linear Form: Reducing y = ax^n to Linear Form (TLMaths)
☐ Gradient of e^kx — Exponential Model: The Gradient of e^(kx) (TLMaths)

Q2.4.01 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 5e^x - 4$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.02 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 35e^{0.3t}$. Find the rate of change of m when $t = 8$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.03 Laws of logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given that $p = \log_3 a$ and $q = \log_3 b$, express $\log_3 \left(\frac{a^3\sqrt{b}}{3b^1}\right)$ in terms of p and q.

(3)

WORKING

Q2.4.04 e^x and ln x ●○○ [2 marks] ☐ Todo ☐ Mastered

Solve $\ln(5x - 4) = 1$, giving your answer in exact form.

(2)

WORKING

Q2.4.05 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 4e^{-x} + 6$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.06 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 32e^{0.3t}$. Find the rate of change of m when $t = 2$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.07 e^x and ln x ●○○ [2 marks] ☐ Todo ☐ Mastered

Solve $\ln(2x - 4) = 1$, giving your answer in exact form.

(2)

WORKING

Q2.4.08 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 1e^{-2x} - 4$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.09 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 11e^{0.05t}$. Find the rate of change of m when $t = 9$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.10 Laws of logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given that $p = \log_3 a$ and $q = \log_3 b$, express $\log_3\left(\frac{a^4\sqrt{b}}{3b^2}\right)$ in terms of p and q.

(3)

WORKING

Q2.4.11 Solving exponential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $2^{x+3} = 3^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.12 e^x and ln x ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $e^{2x} - 9e^x + 18 = 0$, giving your answers as exact values.

(4)

WORKING

Q2.4.13 **Non-linear data: log y against x and log y against log x** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

Two variables are believed to be related by $y = ax^n$. A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line passing through $(0, 1.114)$ with gradient -1. Find a to 2 significant figures and state n.

(3)

WORKING

Q2.4.14 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

The number of rabbits on an island, N, is modelled by $N = Ae^{kt}$, where t is the time in years. Initially $N = 200$, and when $t = 5$, $N = 257$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1000. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.15 **Solving exponential equations** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $3^{x+1} = 7^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.16 **Non-linear data: log y against x and log y against log x** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Two variables are believed to be related by $y = ab^x$. A graph of $\log_{10} y$ against x is a straight line with gradient 0.079 and vertical intercept 1.176. Find the values of a and b to 2 significant figures.

(4)

WORKING

Q2.4.17 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

The number of rabbits on an island, N, is modelled by $N = Ae^{kt}$, where t is the time in years. Initially $N = 800$, and when $t = 6$, $N = 1644$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1600. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.18 **Solving exponential equations** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $3^{x+1} = 5^{x-2}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.19 **Non-linear data: log y against x and log y against log x** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Two variables are believed to be related by $y = ab^x$. A graph of $\log_{10} y$ against x is a straight line with gradient 0.079 and vertical intercept 0.301. Find the values of a and b to 2 significant figures.

(4)

WORKING

Q2.4.20 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The number of bacteria in a culture, N, is modelled by $N = Ae^{kt}$, where t is the time in hours. Initially $N = 400$, and when $t = 9$, $N = 822$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1200. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.21 Solving exponential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $2^{x+3} = 5^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

MY NOTES — 2.4 EXPONENTIALS AND LOGARITHMS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

2.1 Algebraic Methods

Q2.1.01 **Partial fractions · 3 marks**

$\frac{-5}{x+5}+\frac{3}{x-4}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.02 **Algebraic fractions · 3 marks**

$\frac{3x-9}{x+5}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.03 **Proof by exhaustion · 2 marks**

Check $n = 1..6$: $3, 6, 11, 18, 27, 38$; none divisible by 4.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.04 **Disproof by counter-example · 2 marks**

$a = 1$, $b = -2$: $a > b$ but $1 < 4$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.05 **Polynomial division · 4 marks**

quotient $x^2 + 8x + 12$; $(x-1)(x+2)(x+6)$

M1 long division to obtain $xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.06 **Proof by deduction · 3 marks**

Let the integers be n, $n+1$, $n+2$. Sum $= 3n + 3 = 3(n+1)$, a multiple of 3.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.07 **Partial fractions · 3 marks**

$\frac{4}{x-4}-\frac{4}{x+3}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.08 **Algebraic fractions · 3 marks**

$\frac{x-1}{x-3}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.09 **Factor theorem · 2 marks**

$f(1) = 10$, so it is not a factor

M1 evaluates $f(1)$ · A1 value and conclusion

Q2.1.10 **Proof by exhaustion · 4 marks**

Cases $n = 3k$: $9k^2$; $n = 3k \pm 1$: $9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.11 **Polynomial division · 4 marks**

quotient $2x^2 - 8x + 6$; $2(x-4)(x-3)(x-1)$

M1 long division to obtain $2xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.12 **Proof by deduction · 2 marks**

Expands to $12n$, so it is a multiple of 12.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.13 **Partial fractions · 3 marks**

$\frac{2}{x-3}+\frac{5}{x+2}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.14 **Algebraic fractions · 3 marks**

$\frac{x+2}{x-6}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.15 **Factor theorem · 2 marks**

$f(2) = 15$, so it is not a factor

M1 evaluates $f(2)$ · A1 value and conclusion

Q2.1.16 **Polynomial division · 4 marks**

quotient $x^2 - 9x + 20$; $(x-5)(x-4)(x+3)$

M1 long division to obtain $xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.17 **Proof by deduction · 3 marks**

$x^2 - 6x + 11 = (x-3)^2 + 2$; $(x-3)^2 \geq 0$ so the expression $\geq 2 > 0$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.18 **Partial fractions · 3 marks**

$\frac{-3}{x-5} - \frac{1}{x-4}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.19 **Algebraic fractions · 3 marks**

$\frac{2x-8}{x+4}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.20 **Factor theorem · 2 marks**

$f(1) = -10$, so it is not a factor

M1 evaluates $f(1)$ · A1 value and conclusion

Q2.1.21 **Proof by contradiction · 5 marks**

Assume finitely many $p_1, \ldots, p_n$; $N = p_1 p_2 \cdots p_n + 1$ leaves remainder 1 on division by each; so N is prime or has a prime factor not in the list; contradiction.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.22 **Factor theorem · 6 marks**

(a) $p = 52$

M1 sets $f(-2) = 0$ · A1 $p = 52$

(b) $2(x+2)(x+3)(x+4)$

M1 divides by the factor · A1 quadratic factor · M1 factorises · A1

Q2.1.23 **Partial fractions with repeated factors · 4 marks**

$A = -2$, $B = 1$, $C = 1$

M1 multiplies through by the denominator · B1 $B = 1$ (substitute $x = -2$) · B1 $C = 1$ (substitute $x = 3$) · A1 A by comparing x^2 coefficients

Q2.1.24 **Proof by contradiction · 5 marks**

Assume $\sqrt{2} = \frac{a}{b}$ in lowest terms; $a^2 = 2b^2$ so a even, $a = 2c$; then $b^2 = 2c^2$ so b even; contradiction with lowest terms.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.25 **Partial fractions with repeated factors · 4 marks**

$A = 1$, $B = -1$, $C = -2$

M1 multiplies through by the denominator · B1 $B = -1$ (substitute $x = 3$) · B1 $C = -2$ (substitute $x = 1$) · A1 A by comparing x^2 coefficients

Q2.1.26 **Partial fractions with repeated factors · 4 marks**

$A = 5$, $B = 2$, $C = 2$

M1 multiplies through by the denominator · B1 $B = 2$ (substitute $x = -4$) · B1 $C = 2$ (substitute $x = 3$) · A1 A by comparing x^2 coefficients

Q2.1.27 **Proof by contradiction · 4 marks**

Assume n odd, $n = 2k + 1$; $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd; contradiction.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

2.2 Functions and Graphs

Q2.2.01 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, -1)$; y-intercept $(0, 3)$; x-intercepts $(-5, 0)$, $(-3, 0)$

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.02 **Solving modulus equations and inequalities · 3 marks**

$-10 < x < 0$

M1 $-5 < x + 5 < 5$ or equivalent critical values · A1 critical values · A1 correct region

Q2.2.03 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 9$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 9$ · B1 many-to-one

Q2.2.04 **Composite functions · 6 marks**

(a) -48

M1 $g(3) = 23$ · A1

(b) $8x^2 + 16x + 13$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = \frac{-x-2}{2}$

M1 rearranges $y = -2x - 2$ · A1

Q2.2.05 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(1, 1)$

B1 coordinates · B1 transformation described fully

Q2.2.06 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, -3)$; y-intercept $(0, 1)$; x-intercepts $(-7, 0)$, $(-1, 0)$

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.07 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 8$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 8$ · B1 many-to-one

Q2.2.08 **Composite functions · 6 marks**

(a) -5

M1 $g(2) = 0$ · A1

(b) $x^2 + 10x + 21$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = -x - 5$

M1 rearranges $y = -x - 5$ · A1

Q2.2.09 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(2, 4)$

B1 coordinates · B1 transformation described fully

Q2.2.10 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, 5)$; y-intercept $(0, 13)$; no x-intercepts

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.11 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 4$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 4$ · B1 many-to-one

Q2.2.12 **Composite functions · 6 marks**

(a) -29

M1 $g(-3) = 6$ · A1

(b) $25x^2 - 10x - 2$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = \frac{1-x}{5}$

M1 rearranges $y = 1 - 5x$ · A1

Q2.2.13 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(-3, -2)$

B1 coordinates · B1 transformation described fully

Q2.2.14 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**

(a) touches x-axis at 4 and 5; y-intercept 20; maximum at $(\frac{9}{2}, \frac{1}{4})$

B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point

(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -5, -4, 4, 5; y-intercept 20

B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.15 **Inverse functions · 4 marks**

(a) $f^{-1}(x) = \frac{x+3}{3x-4}$

M1 swaps x and y or rearranges · M1 collects y terms · A1

(b) $x \in \mathbb{R}, x \neq \frac{4}{3}$

B1 range of f is all reals except $\frac{4}{3}$

Q2.2.16 **Combining transformations · 3 marks**

translation by $(4, 0)$; vertical stretch scale factor 3; translation by $(0, 2)$

B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

Q2.2.17 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**

(a) touches x-axis at -1 and 3; y-intercept 3; maximum at $(1, 4)$

B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point

(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -3, 3; y-intercept -3

B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.18 **Solving modulus equations and inequalities · 4 marks**

$x = \frac{5}{4}$

M1 solves $2x - 6 = 2x + 1$ · M1 solves $-(2x - 6) = 2x + 1$ · M1 checks the RHS is non-negative / rejects invalid roots · A1 $x = \frac{5}{4}$

Q2.2.19 **Inverse functions · 4 marks**
(a) $f^{-1}(x) = \frac{5x+4}{x+3}$
M1 swaps x and y or rearranges · M1 collects y terms · A1
(b) $x \in \mathbb{R}, x \neq -3$
B1 range of f is all reals except -3

Q2.2.20 **Combining transformations · 3 marks**
translation by $(-2, 0)$; vertical stretch scale factor -1 (including reflection in the x-axis); translation by $(0, 1)$
B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

Q2.2.21 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**
(a) touches x-axis at -3 and 1; y-intercept 3; maximum at $(-1, 4)$
B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point
(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -1, 1; y-intercept -3
B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.22 **Solving modulus equations and inequalities · 4 marks**
$x = -2$
M1 solves $x + 1 = -2x - 3$ · M1 solves $-(x + 1) = -2x - 3$ · M1 checks the RHS is non-negative / rejects invalid roots · A1 $x = -2$

Q2.2.23 **Inverse functions · 4 marks**
(a) $f^{-1}(x) = \frac{3x+1}{3x-3}$
M1 swaps x and y or rearranges · M1 collects y terms · A1
(b) $x \in \mathbb{R}, x \neq 1$
B1 range of f is all reals except 1

Q2.2.24 **Combining transformations · 3 marks**
translation by $(-4, 0)$; vertical stretch scale factor -1 (including reflection in the x-axis); translation by $(0, 2)$
B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

2.3 Binomial Expansion

Q2.3.01 **Range of validity · 1 marks**
$|x| < \frac{7}{4}$
B1

Q2.3.02 **(a + b)^n for positive integer n · 4 marks**
$1 + 28x + 336x^2 + 2240x^3$
B1 first term · M1 correct binomial coefficients with powers of $4x$ · A1 two terms correct · A1 all correct

Q2.3.03 **Range of validity · 1 marks**
$|x| < 3$
B1

Q2.3.04 **(a + b)^n for positive integer n · 4 marks**
$1 - 8x + 24x^2 - 32x^3$
B1 first term · M1 correct binomial coefficients with powers of $-2x$ · A1 two terms correct · A1 all correct

Q2.3.05 **Range of validity · 1 marks**
$|x| < \frac{5}{6}$
B1

Q2.3.06 **(a + b)^n for positive integer n · 4 marks**
$32 + 320x + 1280x^2 + 2560x^3$
B1 first term · M1 correct binomial coefficients with powers of $4x$ · A1 two terms correct · A1 all correct

Q2.3.07 **Range of validity · 1 marks**
$|x| < \frac{3}{5}$
B1

Q2.3.08 **Approximations using expansions · 5 marks**
(a) $1 + 2x - 2x^2$
M1 binomial with $n = \frac{1}{2}$ · A1 · A1
(b) 1.09500
M1 substitutes · A1

Q2.3.09 **(1 + x)^n for rational n · 5 marks**
(a) $1 - 3x + \frac{15x^2}{2} - \frac{35x^3}{2}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < \frac{1}{2}$
B1

Q2.3.10 Approximations using expansions · 5 marks
(a) $1 + x - \frac{x^2}{2}$
M1 binomial with $n = \frac{1}{2}$ · A1 · A1
(b) 1.00995
M1 substitutes · A1

Q2.3.11 (1 + x)^n for rational n · 5 marks
(a) $1 - \frac{x}{6} + \frac{x^2}{18} - \frac{7x^3}{324}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < 2$
B1

Q2.3.12 (1 + x)^n for rational n · 5 marks
(a) $1 + \frac{2x}{3} + \frac{8x^2}{9} + \frac{112x^3}{81}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < \frac{1}{2}$
B1

Q2.3.13 Using partial fractions with expansions · 7 marks
(a) $\frac{1}{1+3x} - \frac{3}{1-3x}$
M1 form · A1 one constant · A1 both
(b) $-2 - 12x - 18x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.14 (a + bx)^n for rational n · 5 marks
$\frac{1}{2} - \frac{x}{16} + \frac{3x^2}{256}$, valid for $|x| < 4$
M1 takes out $4^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

Q2.3.15 Using partial fractions with expansions · 7 marks
(a) $\frac{1}{1+3x} - \frac{3}{1-2x}$
M1 form · A1 one constant · A1 both
(b) $-2 - 9x - 3x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.16 (a + bx)^n for rational n · 5 marks
$\frac{1}{2} + \frac{3x}{16} + \frac{27x^2}{256}$, valid for $|x| < \frac{4}{3}$
M1 takes out $4^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

Q2.3.17 Using partial fractions with expansions · 7 marks
(a) $\frac{-4}{1+3x} - \frac{3}{1-x}$
M1 form · A1 one constant · A1 both
(b) $-7 + 9x - 39x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.18 (a + bx)^n for rational n · 5 marks
$\frac{1}{3} + \frac{x}{18} + \frac{x^2}{72}$, valid for $|x| < 3$
M1 takes out $9^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

2.4 Exponentials and Logarithms

Q2.4.01 Exponential functions and graphs · 3 marks
y-intercept $(0, 1)$; asymptote $y = -4$
B1 shape (increasing) · B1 intercept · B1 asymptote

Q2.4.02 Gradient of e^kx · 2 marks
116 g per unit time
M1 $\frac{dm}{dt} = 10.5e^{0.3t}$ · A1

Q2.4.03 Laws of logarithms · 3 marks
$3p - \frac{1}{2}q - 1$
M1 uses addition/subtraction law · M1 uses power law · A1

Q2.4.04 e^x and ln x · 2 marks
$x = \frac{e^1 + 4}{5}$
M1 $5x - 4 = e^1$ · A1

Q2.4.05 Exponential functions and graphs · 3 marks
y-intercept $(0, 10)$; asymptote $y = 6$
B1 shape (decreasing) · B1 intercept · B1 asymptote

Q2.4.06 **Gradient of e^kx · 2 marks**

17.5 g per unit time

M1 $\frac{dm}{dt} = 9.60e^{0.3t}$ · A1

Q2.4.07 **e^x and ln x · 2 marks**

$x = \frac{e^1 + 4}{2}$

M1 $2x - 4 = e^1$ · A1

Q2.4.08 **Exponential functions and graphs · 3 marks**

y-intercept $(0,\ -3)$; asymptote $y = -4$

B1 shape (decreasing) · B1 intercept · B1 asymptote

Q2.4.09 **Gradient of e^kx · 2 marks**

0.863 g per unit time

M1 $\frac{dm}{dt} = 0.550e^{0.05t}$ · A1

Q2.4.10 **Laws of logarithms · 3 marks**

$4p - \frac{3}{2}q - 1$

M1 uses addition/subtraction law · M1 uses power law · A1

Q2.4.11 **Solving exponential equations · 4 marks**

$x = 13.3$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.12 **e^x and ln x · 4 marks**

$x = \ln 6,\ x = \ln 3$

M1 lets $u = e^x$ · M1 $(u-6)(u-3) = 0$ · M1 takes ln · A1 both exact

Q2.4.13 **Non-linear data: log y against x and log y against log x · 3 marks**

$a = 13,\ n = -1$

M1 $\log y = \log a + n\log x$ · A1 n · A1 a

Q2.4.14 **Exponential growth and decay models · 6 marks**

(a) $k = 0.0502$

B1 $A = 200$ · M1 $257 = 200e^{5k}$ · A1

(b) $t = 32.1$ years

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.15 **Solving exponential equations · 4 marks**

$x = 8.19$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.16 **Non-linear data: log y against x and log y against log x · 4 marks**

$a = 15,\ b = 1.2$

M1 $\log y = \log a + x\log b$ · M1 $\log a = 1.176$ · A1 a · A1 b

Q2.4.17 **Exponential growth and decay models · 6 marks**

(a) $k = 0.120$

B1 $A = 800$ · M1 $1644 = 800e^{6k}$ · A1

(b) $t = 5.77$ years

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.18 **Solving exponential equations · 4 marks**

$x = 8.45$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.19 **Non-linear data: log y against x and log y against log x · 4 marks**

$a = 2.0,\ b = 1.2$

M1 $\log y = \log a + x\log b$ · M1 $\log a = 0.301$ · A1 a · A1 b

Q2.4.20 **Exponential growth and decay models · 6 marks**

(a) $k = 0.0800$

B1 $A = 400$ · M1 $822 = 400e^{9k}$ · A1

(b) $t = 13.7$ hours

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.21 **Solving exponential equations · 4 marks**

$x = 7.54$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1